

Nandini Agrawal

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SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL

Frameworks and Libraries: React, LangGraph, Node.js, Django

Agentic AI and Tool Calling: Claude Workflows, ChatGPT AppsSDK, RAG, LLM Tooling & Orchestration(MCP)

Applications and Tools: AWS, PostgreSQL, Splunk, GitHub, Postman, Selenium, Metamask, Hyperledger Fabric

PROFESSIONAL EXPERIENCE

Intuit

San Diego, CA

Full Stack Software Engineer 2

Feb 2025 - Present

- Built and productionized end-to-end GenAI-powered systems using LangGraph and LLM orchestration, combining React-based frontend workflows with backend orchestration services to power an AI-driven tax filing engine achieving 95% accuracy across 10,000 real customer records.
- Remodeled the sales tax rule extraction for QuickBooks by building an neurosymbolic system that converts complex county tax laws into Clingo based Answer set programming rules, reducing manual rule creation effort by 85% while maintaining 90% accuracy across multiple product categories.
- Architected and implemented a secure Model Context Protocol (MCP) using AppsSDK and LLM tool calling, building authentication, authorization, and API routing layers to safely integrate TurboTax and QuickBooks APIs that led to Intuit launching its GEO initiative within ChatGPT.
- Developed a full-stack AI support automation system, combining RAG pipelines, vector search, backend services, and internal UIs to automate customer support queries and incident root-cause analysis, reducing on-call load by 20–30% during peak periods.

Frontend Software Engineer 1 at TurboTax

Feb 2023 - Jan 2025

- Led modernization of frontend architecture using React and JavaScript, refactoring 5,000+ lines of legacy code into modular, reusable components, reducing production bugs by 50% and improving performance, maintainability, and developer velocity.
- Built and shipped end-to-end product flows using feature flagging, experimentation frameworks, and analytics instrumentation, enabling safe rollout of new SKUs and rapid iteration on business-critical features.
- Optimized performance, reducing load time of a core TurboTax frontend plugin by reducing p95 number from 2s to 1s through improved API design, caching strategies, and rendering optimizations.
- Owned mission-critical production services through on-call rotations, debugging distributed systems issues using Splunk and implementing preventative fixes to improve reliability during peak tax season.

Columbia University

New York, NY

Columbia Data Science Institute Scholar

Jan 2022 – May 2022

- Designed and built both blockchain and non-blockchain implementations of a collaborative prediction market platform used by 50+ students in a graduate-level course with scalable, user-friendly full-stack design.
- Engineered backend services and smart contract incentives using Python, Django, PostgreSQL, and Metamask, enabling secure authentication, precise prediction scoring, and transparent reward distribution.

Ashoka University

Haryana, India

Research and Software Development Intern

Jun 2020 - Aug 2020

- Created [CovidBloc](#), a blockchain-powered contact tracing system combining a Flutter mobile app, Vue web portal, and Node.js backend and Hyperledger Fabric to securely track and store contact events while preserving data.
- Enabled identification and analysis of potential COVID-19 spread patterns across 100+ users within the university campus

Kings College London and Trivedi Center of Political Data

Remote

Software Engineer Intern

May 2020 - Aug 2020

- Built an automated data ingestion and web scraping pipeline for the DecovIndia platform using Selenium to collect and maintain real-time COVID-19 policy notifications from multiple government sources, significantly reducing manual data collection effort.
 - Processed and structured large-scale unstructured text data, enabling downstream policy analysis of health infrastructure, social security programs, and lockdown impacts across Indian states.
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EDUCATION

Columbia University

New York, NY

Master of Science in Computer Science, GPA: 3.79/4.0

Sept 2021 – Dec 2022

Ashoka University

Haryana, India

Postgraduate Diploma in Advanced Studies, Bachelors of Science in Computer Science

Aug 2017 – May 2021

GPA: 3.96/4.0(Diploma), 3.82/4.0(Undergraduate)